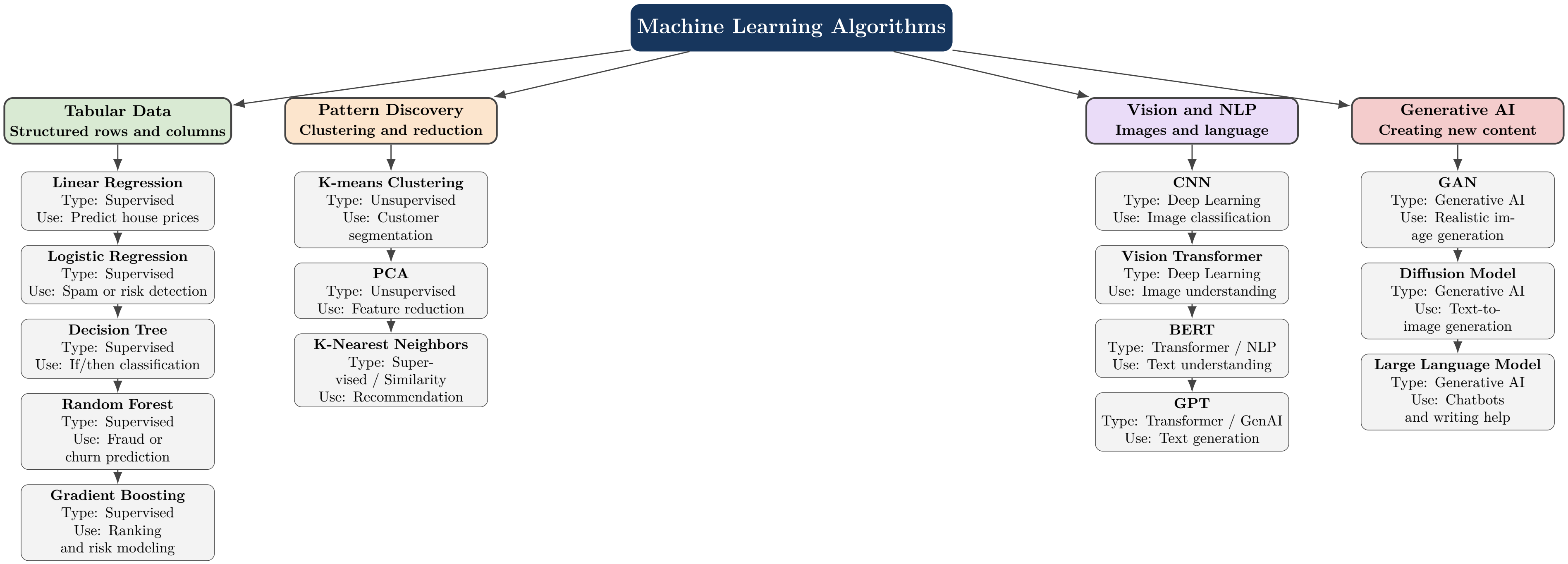
Machine Learning Algorithm Framework

From Tabular Data to Computer Vision, NLP, and Generative AI

AIML Portfolio Artifact — Benita Chinemerem — May 18, 2026



How to Read This Framework

The framework groups algorithms by the kind of problem they usually solve. Supervised algorithms learn from labeled examples, unsupervised algorithms find hidden patterns, deep learning models handle complex data such as images and text, and generative AI models create new content such as text or images.

Category	Common Algorithms	Main Domain	What They Help With
Supervised Learning	Linear Regression, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting	Tabular Data	Predicting values, classifying records, identifying risk, detecting fraud, and supporting decisions.
Unsupervised Learning	K-means, PCA	Tabular Data, Preprocessing	Finding hidden groups, simplifying large datasets, reducing features, and discovering patterns.
Computer Vision and NLP	CNN, Vision Transformer, BERT, GPT	Images and Text	Classifying images, detecting objects, understanding text, summarizing documents, and answering questions.
Generative AI	GAN, Diffusion Model, LLM	Text, Images, Creative AI	Creating new images, generating text, supporting chatbots, producing synthetic data, and assisting creative work.